Introduction to APIs: Wikipedia

Week 10 · APIs module · Textbook Chapter 10

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This week at a glance

We leave scraping behind and start asking servers for data *on purpose* — through a documented API.

Monday | Concepts & notebook intro

- What an API is, why it beats scraping, and why we learn on Wikipedia

Wednesday | Notebook lab

- Query the MediaWiki API: revisions, pageviews, links, raw requests

Friday | Textbook revisions | Proposal due

- Pull requests improving Chapter 10 — **and** your Final Project proposal

Companion notebook

`ch-10-wikipedia.ipynb`

Due Friday, Oct 23

Final Project proposal — research question, data source(s), and your API / scraping plan.

1. Monday · Concepts

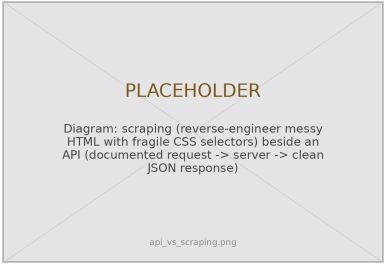
From scraping to APIs

For six weeks we **reverse-engineered** pages built for human eyes — inspecting HTML, guessing selectors, and cleaning up the mess.

This week begins Part III: **API-based data access**. Instead of taking data a site never meant to hand us, we ask for it the way the operator intends.

The skills you build here — reading documentation, constructing parameterized URLs, handling pagination, parsing nested JSON — transfer to *every* API in the rest of the course.

When an API exists, it is almost always the better tool.



Ask, don't reverse-engineer.

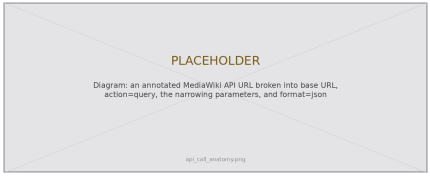
Textbook Ch. 10, "From Document Scraping to API Access."

What is an API?

An **API** (Application Programming Interface) is a **contract**: it specifies what you may request, what parameters you may pass, and what format the response will take.

- Clean **JSON**, not cluttered HTML
- **Stable endpoints**, not fragile CSS selectors
- **Documentation** tells you exactly what exists

The tradeoff: an API only exposes what its operator *chooses* to. You cannot query a missing field, exceed the rate limits, or count on it forever — the "post-API age" (later this module) is a graveyard of APIs that were restricted, repriced, or shut down.



A request is a URL you assemble on purpose.

Textbook Ch. 10, "What Is an API?"

Why Wikipedia?

The **MediaWiki API** is a near-perfect place to learn:

- Free, with no authentication beyond a **User-Agent** header
- Comprehensive, well-maintained documentation
- Data rich enough for real research — content, networks, events

Our case study: the `Elizabeth_II` article. Its editing history has dramatic spikes around major events — ideal for seeing temporal patterns in API data.

Every edit, ever, is queryable.

Public interest

Wikipedia is the counter-example to enclosure: an **open API**, an **open license**, and public database **dumps**. Nothing you learn here sits at the mercy of a pricing change.

Textbook Ch. 10, "Why Wikipedia?" & the "Public Interest Connection" callout.

The anatomy of a request — and what the notebook covers

Every MediaWiki call is four parts:

1. a **base URL** — `.../w/api.php`
2. an `action` — what you want to do
3. **parameters** that narrow the request
4. a `format` — `json`

The response is nested JSON you navigate **one level at a time**: `data["query"]["pages"][page_id]`. Internalize that shape once and every endpoint feels familiar.

This week's notebook

- Revisions + pagination
- Pageviews (a second API)
- Links, categories, languages
- Raw requests from the docs

Connecting to Ch. 2 (Ethics)

Identify yourself: the Wikimedia **User-Agent policy** asks every client for a descriptive agent with contact info. Anonymous traffic gets throttled.

Textbook Ch. 10, "The Anatomy of an API Call." See also *Missing Manual* Ch. 5, "Reading Official Documentation."

2. Wednesday · Notebook Lab

The four parts, then navigate the JSON

Assemble the request as a dictionary of parameters; `requests` builds the URL for you:

```
import requests
URL = "https://en.wikipedia.org/w/api.php"
HEADERS = {"User-Agent": "WebDataScience/1.0 (INFO 4617; brian.keegan@colorado.edu)"}

params = {"action": "query", # what to do
          "titles": "Elizabeth_II", # about this page
          "prop": "info", # what we want back
          "format": "json"} # how we want it
data = requests.get(URL, params=params, headers=HEADERS).json()
```

Then walk the nested response one key at a time — page IDs are the keys:

```
pages = data["query"]["pages"]
page_id = list(pages.keys())[0]
print(pages[page_id]["title"], pages[page_id]["pageid"])
```

Textbook Ch. 10, "The Anatomy of an API Call." Define `HEADERS` once and reuse it everywhere.

Revisions — and the pagination loop

```python
def get_revisions(title, headers=HEADERS):
    params = {"action": "query", "titles": title,
        "prop": "revisions", "rvlimit": "max",
        "rvprop": "ids|timestamp|user|size|comment",
        "format": "json"}
    revs = []
    while True: # the pagination loop
        data = requests.get(URL, params=params,
                            headers=headers).json()
        page = list(data["query"]["pages"].values())[0]
        revs.extend(page.get("revisions", []))
        if "continue" not in data: # last batch -> stop
            break
        params.update(data["continue"]) # else: next batch
    return pd.DataFrame(revs)
```

The single most common API bug: **forgetting to paginate**. The server returns results in batches (here, capped at 500) and hands back a `continue` token when more remain.

Skip the loop and you silently keep a *fraction* of the data.

Two habits: `raise_for_status()` to fail loudly, and `maxlag=5` so bulk jobs yield to human readers.

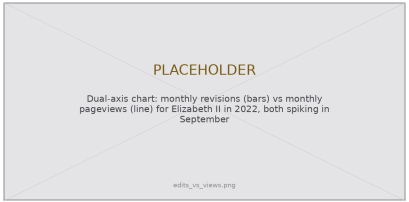
Textbook Ch. 10, “Getting Revisions” & “Understanding Pagination.” The same loop returns in the government & social chapters.

Same pattern, more endpoints

Pageviews live on a **different** API (REST) — parameters are baked into the URL *path*, and there is no continuation:

```
def get_pageviews(title, start, end, headers=HEADERS):
    url = ("https://wikimedia.org/api/rest_v1/metrics"
           "/pageviews/per-article/en.wikipedia"
           f"/all-access/all-agents/{title}"
           f"/daily/{start}/{end}")
    r = requests.get(url, headers=headers)
    r.raise_for_status()
    return pd.DataFrame(r.json()["items"])
```

Links, categories, and languages reuse the *same* Action-API loop — just swap `prop` and the matching `*limit` key (`links`, `categories`, `langlinks`).



Edits vs. views, 2022 — both spike in September.

How many of Wikipedia's 300+ languages carry an article is a rough proxy for **global significance** — one that skews toward the Global North.

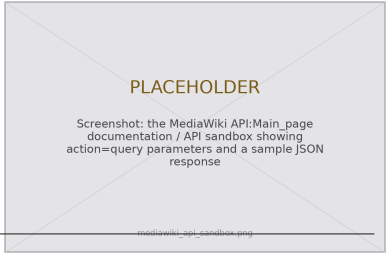
Textbook Ch. 10, "Getting Pageviews," "Getting Links and Categories," "Interlanguage Links."

From documentation to working code

The transferable skill is writing a request from the docs alone — no wrapper. Read the parameters for an endpoint you have never used, build it, inspect the response:

```
params = {"action": "query",
          "list": "search", # a NEW endpoint, learned from the docs
          "srsearch": "web scraping",
          "srlimit": 10,
          "srprop": "wordcount|timestamp|snippet",
          "format": "json"}
data = requests.get(URL, params=params, headers=HEADERS).json()
for r in data["query"]["search"]:
    print(r["title"], "-", r["wordcount"], "words")
```

Libraries like `mwclient` and `pywikibot` automate continuation and throttling — but we write raw requests first so you know exactly what they do for you.



Textbook Ch. 10, “Writing Raw API Requests from Documentation.”

Lab: work these, then show-and-tell

Work through the chapter's exercises in pairs:

1. Compare revision histories of two related articles (unique editors, edits-per-editor, monthly activity)
2. Detect events: daily pageviews for five articles — did they spike together or separately?
3. Link network: overlap of outgoing wikilinks across three related articles
4. Raw API practice: build a request for a new endpoint (`extlinks`, `recentchanges`, `redirects`) from the docs
5. Editor network: top-20 editors of an article, then what *else* they edit
6. 5617: reproduce a measure from Mesgari et al. (2015) — e.g. a Gini coefficient of edits per editor

Show-and-Tell

Come ready to share:

- a challenging bug
- interesting data you pulled
- a provocation for a research design

Scraping asks: *how is this page built?*
An API asks: *what does the documentation promise?*

Learn one API by hand,
and you have learned them all.

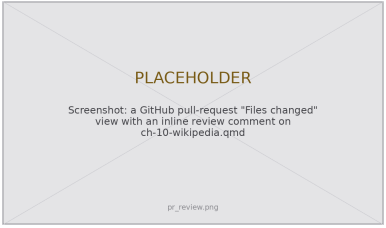
3. Friday · Textbook Revisions

Improving Chapter 10 together

Today we make the book better. Each of you opens a **pull request** against `ch-10-wikipedia.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



Also due today — Fri, Oct 23

Submit your **Final Project proposal**: a research question, your data source(s), and whether you will use an API, scraping, or both.

Contributions & reviews count toward the Textbook Revisions grade (30%).

A revision menu for Chapter 10

High-value targets — pick one and scope it to a single PR:

- **Close a gap in "Common Issues to Debug."** Add a worked **missing-page** check (a bad title returns page ID `-1`), or a `urllib.parse.quote()` example for titles with parentheses / accents.
- **Show the etiquette, don't just name it.** The chapter mentions `maxlag` but never demonstrates it — add a small retry-on-`maxlag` example for bulk jobs.
- **Add a figure.** A diagram of the request anatomy, or of the request → `continue` → request pagination loop, would anchor the two hardest ideas.
- **Strengthen an exercise.** Exercises 1–5 have no expected output or starter cell — add a rubric or a self-checking assertion.
- **Deepen "Social History."** Turn the gender-gap discussion into a reproducible measurement (tie it to WikiProject Women in Red) using this chapter's helper functions.

Targets drawn from the chapter's "Common Issues," callouts, thin exercises, and "Further Reading."

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Government APIs —** the Census, the Federal Reserve (FRED), and other public data services.

Key takeaways

1. An API is a **documented contract**: structured, intentional access — clean JSON and stable endpoints, no reverse-engineering.
2. The skill transfers: **read the docs** → **build a parameterized request** → **parse nested JSON** one level at a time.
3. Always **paginate**: follow the `continue` token in a loop, or you silently keep only the first batch.
4. Be a good client: a descriptive **User-Agent**, serial requests, and `maxlag` for bulk jobs (the ethics of Ch. 2, applied).
5. Openness is a *choice*, not a guarantee of completeness — Wikipedia's data still inherits its volunteer community's biases.

Reading: Textbook Ch. 10. Further: Ford (2022); Mesgari et al. (2015); Salganik (2018).